DATA → DECISIONS → INTERACTIVE RECOMMENDATIONS

Fiber Intake Explorer

From nutrient composition data to practical food combinations

Audience: people exploring food combinations to reach a user-selected fiber goal.

Core question: Which 2–3 foods can meet a fiber target, optionally within a total weight limit?

Yixiao Xiong • Michigan State University • October 22, 2026

01 Data source & reproducible preparation

5,993 food entries

CNF 2026 relational database

Food_Name + Nutrient_Amount + Nutrient_Name + CNF_Food_Group

1,353 in scope

Fruit, Vegetable, Legume

Food_Code identifies each food; nutrient values refer to 100 g edible portion.

1,240 cleaned

113 missing fiber entries excluded

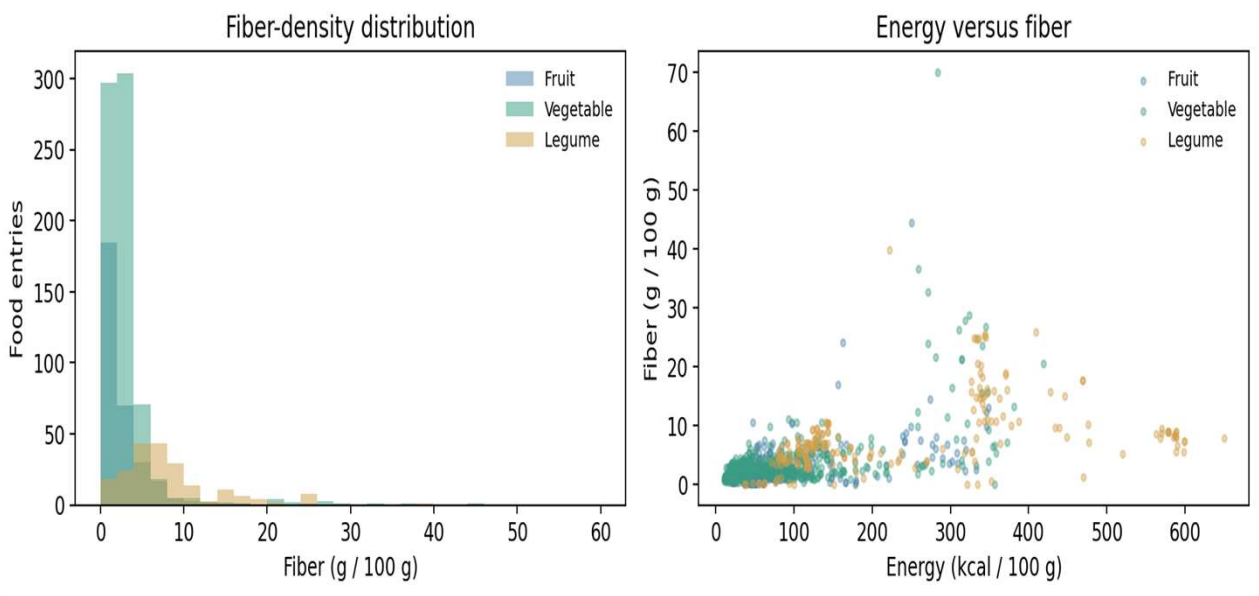
Missing ≠ zero; no unsupported imputation.

Data-quality decisions

Join by documented keys • assess missingness, numeric ranges and duplicates • retain recorded zero values • standardize fiber (g/100 g) and energy (kcal/100 g).

Source: Health Canada, Canadian Nutrient File 2026; official database structure and users' guide.

02 EDA: nutritious does not always mean practical



Top five highest-fiber entries

Cloud ear mushroom, dried	70.1
Baobab powder	44.5
Carob flour	39.8
Seaweed, dried	36.7
Parsley, freeze-dried	32.7

Fiber content: g per 100 g

Finding: extreme fiber density often reflects dried, powdered or minor-use ingredients—not realistic 100 g servings.

This motivates a separate, rule-based suitability screening before recommendation.

03 Feature engineering: make recommendations usable

1 INPUT
1,240 cleaned CNF foods
Fiber and energy per 100 g

↓

2 CLASSIFY
Parse food descriptions
Raw / Cooked / Dried / Processed / Other

↓

3 SCREEN
Flag herbs, spices, powders, peels
and unsuitable dry ingredients

↓

4 OUTPUT
1,048 eligible candidates
192 excluded from default search

Exception: edible dried fruit may remain; untreated dry corn and dehydrated vegetables are screened out.

Rules are transparent heuristics, not measured consumer popularity or validated suitability labels.

04 Live demo: the interactive data product

User inputs

1. Desired fiber (g)
2. Optional maximum combined weight
3. Food groups
4. Two or three foods
5. Optional energy summary

Recommendation workflow

Generate feasible combinations at tested portions (50–200 g)

Click any row to inspect combined weight, fiber and constituent foods.

Rank by practicality then weight and fiber excess.

DEMO: 15 g fiber • all groups • 3 foods • optionally enable a weight cap

05 Interpretation, limits & reproducibility

What the app can say

Uses CNF values to calculate fiber for specified portions and identify feasible combinations within tested candidates.

Its purpose is exploration, not a complete dietary plan.

What it cannot claim

Not globally optimal • not a population popularity ranking • simplified portions • heuristic food screening • nutrition varies by preparation • excludes prices and individual dietary needs.

GitHub: app.py + food_features.py + cleaned CSV + reproducible EDA notebook + README / setup / AI disclosure

Next step: refine food-suitability screening and use realistic portion conversions.